

# Week 7 — Hypothesis testing II & A/B testing

Business Forecasting

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# Where we are

Describing data → **Reasoning under uncertainty** → Regression → Time series forecasting

**Last week of inference.** Last week we tested a claim about *one* group. This week we **compare two groups** — and design the experiment that makes the comparison fair.

- **This week (Testing II): two-sample & paired tests, correlation, and A/B testing** — the causal-inference tool behind modern business experiments.

**Fast & hands-on:** every test you run in R (`t.test`, `prop.test`, `cor.test`) — the exercises appear **as slides** in this deck and the code runs right here in the browser.



# Before we start — two minutes on last week

`Sample_list` holds the 200 sampled Mexico City listings, with `price` the nightly price in pesos. A pricing report claims the true mean nightly price is **1000**.

1. Compute the mean and the standard deviation of `Sample_list$price`, then the standard error of the mean (the sample size is 200).
2. Build the one-sample test statistic for  $H_0 : \mu = 1000$ .
3. Turn it into a two-sided p-value with `pt()` at 199 degrees of freedom.

Exercise

Start Over

Show Hint

Run Code

```
1 # task 1: xbar, s, and the standard error
2
3 # task 2: the test statistic against 1000
4
5 # task 3: the two-sided p-value
```

# Hypothesis Testing: Two Samples

# Two samples and the difference in means

Two iid samples, **independent** of each other, from two populations  $X_1, \dots, X_n$  and  $Y_1, \dots, Y_m$ .

**General idea:** test the difference between **population means**.

- Ex:  $H_0 : \mu_X - \mu_Y = 100$  vs  $H_A : \mu_X - \mu_Y \neq 100$
- Ex:  $H_0 : \mu_X - \mu_Y = 0$  vs  $H_A : \mu_X - \mu_Y > 0$
- Test for no difference:  $H_0 : \mu_X - \mu_Y = 0$  vs  $H_A : \mu_X - \mu_Y \neq 0$
- **Test statistic:** normalized difference in sample means

$$\text{test stat} = \frac{\bar{X} - \bar{Y} - \overbrace{(\mu_{X,0} - \mu_{Y,0})}^{\Delta_0}}{SE} = \frac{\bar{X} - \bar{Y} - \Delta_0}{SE}$$

# Variance in two samples

- SE depends on the estimate of the standard deviation:

- If standard deviation known:  $SE = \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$

- Why? Because  $Var(\bar{X} - \bar{Y}) = Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$

- If standard deviation not known:  $SE = \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$

- **Rejection region:** depends on the distribution

- Large sample or known variance from normal:  $Z_{test} \sim N(0, 1)$

- Small sample with normal data:  $T_{test} \sim t(v)$  — critical values from Student's t,  $v$  df

# Side note — degrees of freedom

Degrees of freedom for a difference in means. Two ways to do it:

1. Proper, complex (and annoying):

$$v = \frac{\left( \frac{s_X^2}{n} + \frac{s_Y^2}{m} \right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

Round down to the nearest integer.

2. Simple, approximate shortcut:

$$v = \min(n - 1, m - 1)$$

# Difference in means — two-sided test

Hypothesis:  $H_0 : \mu_X - \mu_Y = \Delta_0$  and  $H_A : \mu_X - \mu_Y \neq \Delta_0$

Rejection Region: for a test of significance level  $\alpha$ , reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \bar{Y} - \Delta_0}{SE} < z_{\frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{X} - \bar{Y} - \Delta_0}{SE} > z_{1-\frac{\alpha}{2}}$$

- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - \Delta_0}{SE} < t_{v, \frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{X} - \bar{Y} - \Delta_0}{SE} > t_{v, 1-\frac{\alpha}{2}}$$

Where  $z_{1-\frac{\alpha}{2}}$  and  $t_{v, 1-\frac{\alpha}{2}}$  are  $(1 - \frac{\alpha}{2})$  quantiles of the standard normal and Student's t with  $v$  df.

# Two-sided test under the CLT

If CLT kicks in, the distributions are:

$$\bar{X} \sim N(\mu_X, \sigma_X/\sqrt{n}) \quad \text{and} \quad \bar{Y} \sim N(\mu_Y, \sigma_Y/\sqrt{m})$$

The difference of two independent normals:

$$\bar{X} - \bar{Y} \sim N(\mu_X - \mu_Y, \sqrt{\sigma_X^2/n + \sigma_Y^2/m})$$

So if the null is true, the standardized statistic:

$$\frac{\bar{X} - \bar{Y} - (\mu_{X,0} - \mu_{Y,0})}{\sqrt{s_X^2/n + s_Y^2/m}} \sim N(0, 1)$$

# Where exactly do you cut?

The rule says “reject in the tails”. The tails have to be *somewhere*: two numbers on the horizontal axis, chosen so the shaded area equals  $\alpha$ .

For  $\alpha = 0.05$ , two-sided: what are those two numbers under  $N(0, 1)$ ?

And if the sample is small so the statistic follows  $t(v)$  instead — do the cutoffs move **in** or **out**?



# Exercise 1 — Draw the rejection region

1. Compute the two cutoffs that leave  $\alpha/2 = 0.025$  in each tail of  $N(0, 1)$ .
2. Draw the density of the test statistic under  $H_0$  and mark both cutoffs on it.
3. Compute the same cutoff from Student's t with  $v = 5, 20, 100, 1000$  degrees of freedom, and say which way they move as  $v$  grows.

Exercise

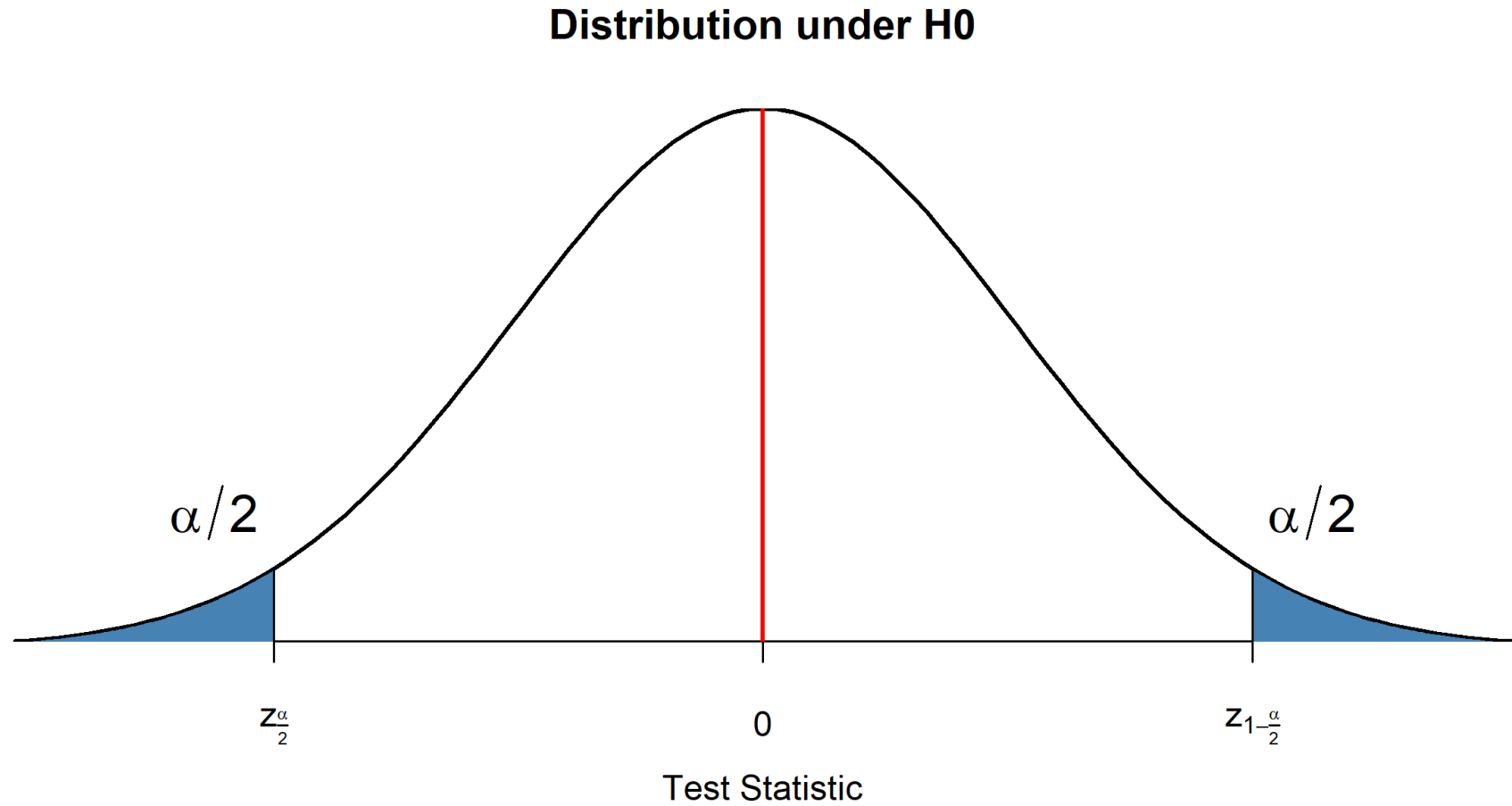
↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

# Two-sided rejection region





# Warm-up — meet the experiment data

The exercises use a households environment survey ([exam\\_data](#)). Two columns matter most:

- [gasoline\\_spending\\_weekly](#) — weekly gasoline spending (the outcome).
- [uses\\_electricity\\_cooling](#) — logical grouping variable ([TRUE/FALSE](#)).

Run this to see the two groups. **Press Run — R runs in your browser.**

Exercise

↺ Start Over

⚠ Show Solution

▶ Run Code

```
1 table(exam_data$uses_electricity_cooling)
2 tapply(exam_data$gasoline_spending_weekly,
3        exam_data$uses_electricity_cooling, mean)
```

## Exercise 2 — Two-sample difference in means

Run a two-sample (Welch) t-test comparing weekly gasoline spending between households that do and do not use electricity for cooling. Read off the two group means and the difference between them.

Exercise

 Start Over

 Show Hint

 Run Code

1 # Your code here

Which group spends more on gasoline per week, on average?

**Households that use electricity cooling (TRUE)** — about 255 per week against about 152 for the **FALSE** group, a gap of roughly 102.

# Confidence interval for a difference

Following this idea, we can construct a confidence interval for the difference:

$$P\left(\underbrace{z_{\alpha/2}} < \frac{\bar{X} - \bar{Y} - (\mu_{X,0} - \mu_{Y,0})}{\sqrt{s_X^2/n + s_Y^2/m}} < \underbrace{z_{1-\alpha/2}}\right) = \underbrace{1 - \alpha}$$
$$CI_{1-\alpha} = \left(\underbrace{\bar{X} - \bar{Y}} \pm \underbrace{z_{1-\alpha/2} \sqrt{s_X^2/n + s_Y^2/m}}\right)$$

## Exercise 3 — P-value and confidence interval

Run the same test again, but this time **save the test object**, then extract **only** the p-value and the 95% confidence interval for the difference in means using base R accessors.

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # Your code here
```

Does the 95% CI for the difference contain 0?

**No** — both endpoints are negative (roughly  $-115$  to  $-89$ ), so 0 is excluded.



## Exercise 4 — Decision at $\alpha = 0.05$

Look at the two numbers you just produced in Exercise 3 — nothing else.

At  $\alpha = 0.05$ , what is the correct decision for  $H_0$ : the two group means are equal?

**Reject  $H_0$**  — the p-value came out far below 0.05, and the CI excludes 0.

The rule: reject  $H_0$  when  $p < \alpha$  — equivalently, when the CI excludes  $\Delta_0$ . (Rejecting only “if we first assume equal variances” is wrong — Welch’s test needs no such assumption.)

# Two samples — mean — one-sided — case 1

**Hypothesis:**  $H_0 : \mu_X - \mu_Y = \Delta_0$  and  $H_A : \mu_X - \mu_Y < \Delta_0$

- This includes any null like  $\mu_X - \mu_Y \geq \Delta_0$ . Why?
- If you rejected  $\mu_X - \mu_Y = \Delta_0$  at  $\alpha$ , you would reject for sure anything larger than  $\Delta_0$ .

**Rejection Region:** for a test of significance level  $\alpha$ , reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \bar{Y} - \Delta_0}{\sigma/\sqrt{n}} < z_\alpha$$

- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} < t_{v,\alpha}$$

Where  $z_\alpha$  and  $t_{v,\alpha}$  are  $\alpha$  quantiles of the standard normal and Student's t with  $v$  df.

# The whole $\alpha$ goes in one tail

Two-sided at  $\alpha = 0.05$  splits the 5% into two tails of 2.5%. One-sided puts all 5% into a single tail.

Write down your prediction before running anything: does the one-sided cutoff sit **further from zero** or **closer to zero** than 1.96?

And what does that do to the p-value of a statistic that points in the alternative's direction?



## Exercise 5 — One-sided cutoffs

Your prediction: the one-sided cutoff is \_\_\_\_\_ than 1.96.

Now produce the evidence.

1. Compute the two-sided cutoff and the one-sided cutoff at  $\alpha = 0.05$ .
2. Take a test statistic of  $-2.10$ . Compute its two-sided p-value and its one-sided (“less”) p-value under  $N(0, 1)$ .
3. State the exact arithmetic relationship between those two p-values.

Exercise

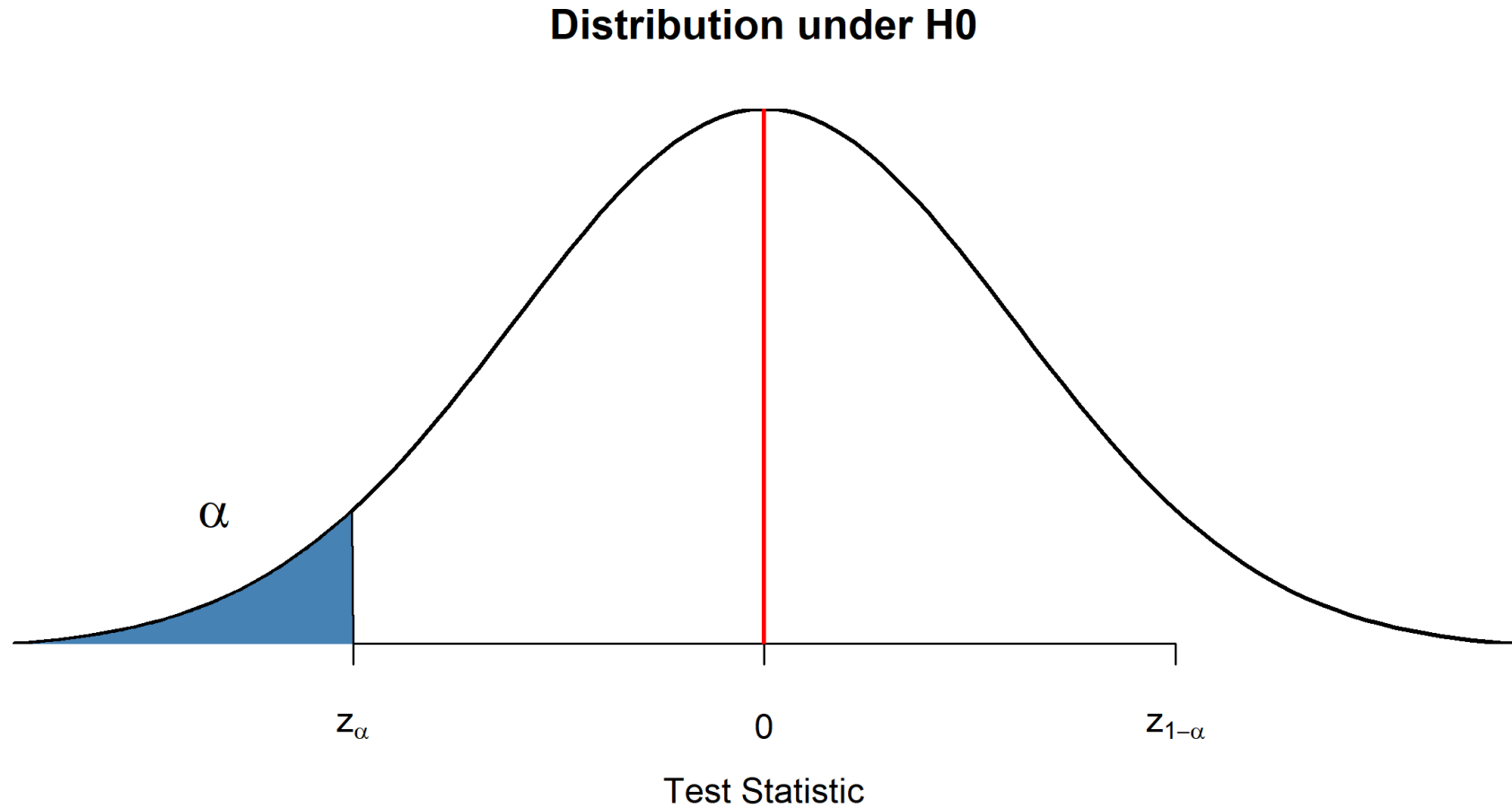
↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

# One-sided rejection region — case 1



# Two samples — mean — one-sided — case 2

**Hypothesis:**  $H_0 : \mu_X - \mu_Y = \Delta_0$  and  $H_A : \mu_X - \mu_Y > \Delta_0$

- This includes any null like  $\mu_X - \mu_Y \leq \Delta_0$ . Why?
- If you rejected  $\mu_X - \mu_Y = \Delta_0$  at  $\alpha$ , you would reject for sure anything smaller than  $\Delta_0$ .

**Rejection Region:** for a test of significance level  $\alpha$ , reject

- If large sample or known variance with normal

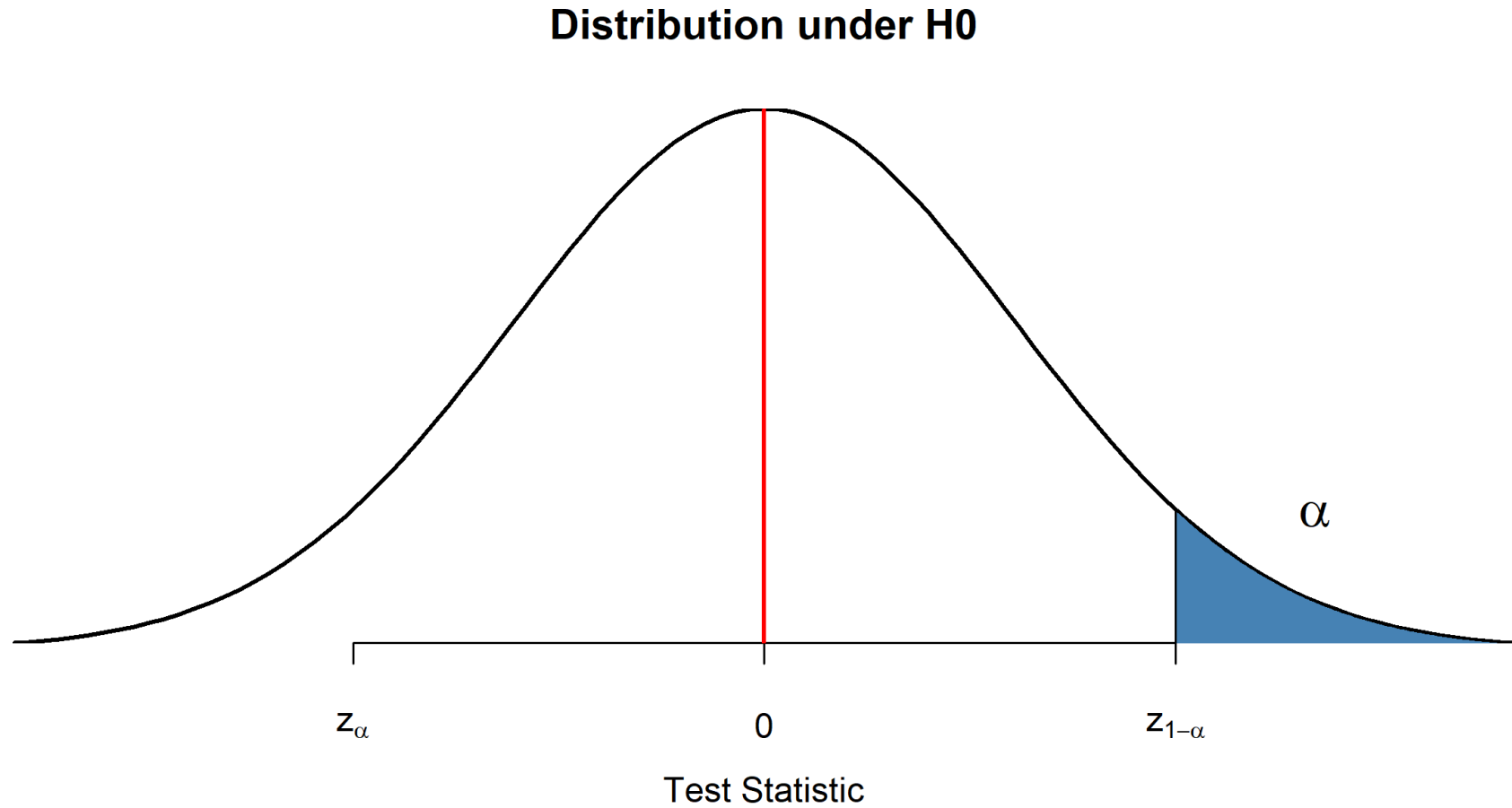
$$\frac{\bar{X} - \bar{Y} - \Delta_0}{SE} > z_{1-\alpha}$$

- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} > \underline{t}_{v,1-\alpha}$$

Where  $z_{1-\alpha}$  and  $t_{v,1-\alpha}$  are  $(1 - \alpha)$  quantiles of the standard normal and Student's t with  $v$  df.

# One-sided rejection region — case 2





## Exercise 6 — One-sided two-sample test

Suppose the claim is that households **without** electricity cooling (**FALSE**) spend *less* on gasoline than those **with** it (**TRUE**). Because R orders the groups **FALSE** then **TRUE**, the difference it reports is (FALSE – TRUE). Run the matching **one-sided** two-sample test, and compare its p-value with the two-sided one from Exercise 2.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

# Airbnb: clean vs dirty prices

You run Airbnbs in Mexico City. You have a sample of 200 listings, split into **Clean** (cleanliness score above 4.5) and **Dirty**.

The business question: **can you charge more if the listing is clean?**

Before any test: what do the two groups actually look like — how many listings, what average price, how spread out?



## Exercise 7 — Summarise the two groups

`Sample_list` holds the 200 listings: `price` and the logical `clean` (and a `clean_label` version reading “Clean” / “Dirty”).

1. For each group, report the **number of listings**, the **mean price** and the **standard deviation** — in one table.
2. Draw the picture that lets you compare the two price distributions at a glance.
3. Say which of the two groups has the larger *spread*, and whether that surprises you.

Exercise

↺ Start Over

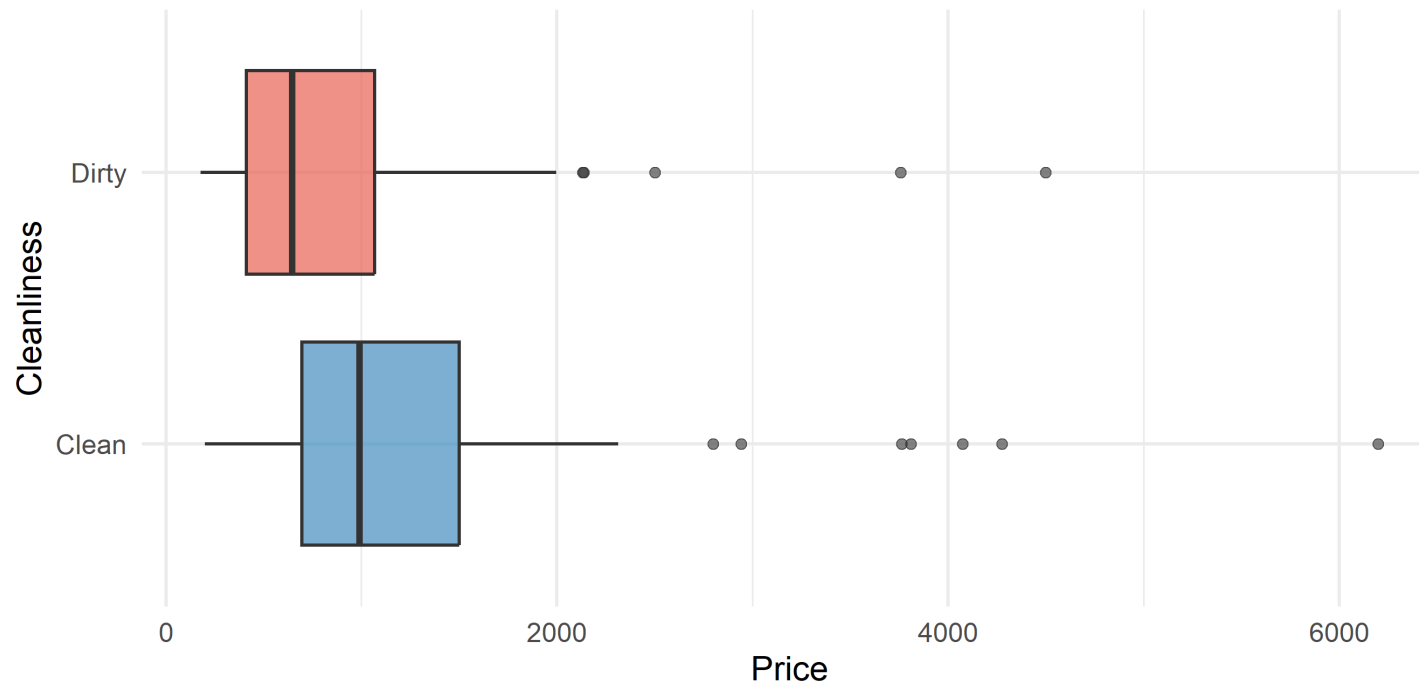
💡 Show Hint

▶ Run Code

```
1 # Your code here
```

# Airbnb: clean vs dirty — the summary

|       | n   | Mean   | SD    |
|-------|-----|--------|-------|
| Clean | 100 | 1245.4 | 961.9 |
| Dirty | 100 | 868.7  | 693.3 |



- State the null and alternative hypothesis
- Calculate the value of the test statistic
- Calculate the p-value; at which  $\alpha$  do we reject? Build the 95% CI for the difference.



## Exercise 8 — The test by hand, then by button

Use **only** the six summary numbers from the table above — no raw data — to test  $H_0 : \mu_{clean} - \mu_{dirty} = 0$  against a two-sided alternative.

1. The standard error of the difference.
2. The test statistic.
3. The degrees of freedom, **both** ways: the full Welch formula from the “Side note” slide, and the shortcut  $v = \min(n - 1, m - 1)$ .
4. The two-sided p-value and the 95% CI.
5. Now check yourself: run `t.test()` on the raw data, and run it again with `var.equal = TRUE`. Compare the `df` line in each output with what you computed.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n_c <- 100; mean_c <- 1245.43; sd_c <- 961.90 # clean
2 n_d <- 100; mean_d <- 868.70; sd_d <- 693.31 # dirty
3
4 # 1-4: your arithmetic here
5
6 # 5: the two R calls that check it
```

# Pooled or Welch?

- **Pooled** ( $v = n + m - 2$ ) assumes  $\sigma_X = \sigma_Y$  and averages the two sample variances into one. Cheap, and wrong when the spreads differ.
- **Welch** makes no such assumption, pays for it with a **fractional**  $v$ , and is what `t.test()` does by default.
- The exam shortcut  $v = \min(n - 1, m - 1)$  is deliberately conservative: fewer df  $\rightarrow$  fatter tails  $\rightarrow$  you reject slightly less often than you should. Safe, never anti-conservative.

# Exercise: McDonald's vs Burger King

We suspect Burger King has larger caloric content for kids' meals.

| Sample      | n  | Sample Mean | Sample Standard Deviation |
|-------------|----|-------------|---------------------------|
| McDonald    | 9  | 475         | 230                       |
| Burger King | 16 | 520         | 180                       |

- State the null and alternative hypothesis
- Calculate the value of the test statistic
- Determine the appropriate number of degrees of freedom
- Calculate the p-value; at which  $\alpha$  do we reject?

## Exercise 9 — Small samples: does the df choice matter?

Same job as Exercise 8, but now  $n = 9$  and  $m = 16$  — small enough that the t distribution really bites.

1. Write  $H_0$  and  $H_A$  for the claim “Burger King meals have **more** calories”.
2. Compute the SE and the test statistic.
3. Compute both degrees of freedom: the Welch formula and  $\min(n - 1, m - 1)$ .
4. Compute the **one-sided** p-value under each. How far apart are they?
5. At  $\alpha = 0.05$ , does the choice of df change your decision?

Exercise

 Start Over

 Show Hint

 Run Code

```
1 n_m <- 9; mean_m <- 475; sd_m <- 230 # McDonald's
2 n_b <- 16; mean_b <- 520; sd_b <- 180 # Burger King
3
4 # your code here
```

# A/B testing: the business question



- **Airbnb** wants to know: *Do professional photos increase host revenue?*
- **Historical data:** listings with better photos show higher revenues.
- But... **Correlation  $\neq$  Causation.** Better photos may be linked to other factors: more organized hosts, higher-quality listings, better service.
- A simple comparison only tells us about **association**, not causality.

# The business decision

- Should Airbnb **invest in professional photography** for hosts?
- If photos aren't the true driver → the investment wastes money. We need a method to **isolate the effect of photos**.

## Enter A/B Testing

### Definition:

- A method to measure *causal* effects.
- Widely used in modern business, especially tech.

### Examples:

- Netflix: which thumbnail gets more clicks?
- Amazon: does a new checkout button increase sales?
- Instagram: do longer videos increase engagement?

# The setup

We divide hosts into **two groups**:

1. **Control group**: no intervention (existing photos).
2. **Treatment group**: Airbnb provides **professional photography**.

**The KEY**: groups are assigned **randomly**.



## Exercise 10 — What makes an A/B test valid

Four candidate answers. Which single feature is what lets an A/B test measure a **causal** effect?

- a large enough sample in each arm
- running the test long enough to see the effect
- assigning units to treatment and control **at random**
- measuring an outcome that management cares about

**Random assignment.** It is the only one of the four that makes the two groups comparable on *everything you did not measure*.

Sample size and duration buy **precision** — a narrower CI around whatever you estimate. Randomization buys **validity** — the guarantee that what you estimate is the treatment effect and not a difference between the kinds of units in each arm.

# Measuring the effect

- After randomization: **the only difference is the photos.**
- Provide photography to the treatment group
- Wait to see potential effects
- Collect outcome data (revenue, reviews, bookings)
- Run a simple two-sample test between treatment and control
- If the treatment group earns more revenue, we know it was **caused by the professional photos**

**This is the causal effect.**

- A/B testing is used extensively in modern business, especially in tech.
- We have all been subjects of these experiments, often without knowing.



## Exercise 11 — A/B test as a proportion test

An A/B test on a checkout button. The **control** group saw the old button; the **treatment** group saw the new one. Conversions (purchases) are:

| Group     | Conversions | Visitors |
|-----------|-------------|----------|
| Control   | 120         | 2000     |
| Treatment | 155         | 2000     |

The outcome here is binary, so the two-sample test is a test of **proportions**. Run it with `prop.test()`, and also compute the two conversion rates yourself.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

1 # Your code here



## Exercise 12 — Reading the proportion test

Reuse the A/B test from Exercise 11. Store the result, extract the **p-value** and the **confidence interval for the difference in proportions**, and reach a decision at  $\alpha = 0.05$ .

Then rebuild the same z-statistic by hand from the pooled proportion, and check that the two routes agree.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

# Does randomization really balance the groups?

The claim is strong: split units at random and the two arms end up identical on *everything* — household size, income, region, and every variable you never even recorded.

That is a claim you can check. Take units that were never treated at all, split them at random, and compare their characteristics across the fake “arms”.

If the claim is right, what should the comparison show?



## Exercise 13 — Check the randomization

Use `exam_data` (14,505 households). Nobody is treated — you are only going to *pretend* to assign them.

1. Assign every household at random to group A or group B.
2. Compare the two groups on `household_size`, `electricity_bill_monthly` and `num_vehicles`, and on the categorical `locality_size`.
3. Run a two-sample t-test on one of those characteristics. What should its p-value look like, and what does it look like?
4. Re-run the whole thing a few times. Does any single balance test ever come out “significant”?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

# Why randomization works

- Both arms are two samples from the **same** population — so every expectation, observed or not, is the same in both.
- Therefore any post-treatment difference in the outcome can only come from the treatment.
- **This is what gives us causal inference.**

Without randomization the groups differ in ways you cannot list, let alone control for.

# If the button does nothing, how often will you still “find” an effect?

Your team ships an A/B test on a button that — unknown to everyone — changes nothing at all. True conversion rate: 6% in both arms.

You run the test on 2,000 visitors per arm and look at the p-value.

Write down your guess: out of 1,000 such tests on a *useless* button, how many come back significant at  $\alpha = 0.05$ ?

And what shape does the histogram of those 1,000 p-values have?



## Exercise 14 — An A/B test with no effect

Your guess: \_\_\_\_\_ out of 1,000.

Simulate it. One “experiment” = draw conversions for both arms from the *same* 6% rate, then test.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # One experiment with NO true effect: both arms convert at 6%.
2 ✓ one_test <- function(n = 2000, p_control = 0.06, p_treat = 0.06) {
3   xC <- rbinom(1, n, p_control)
4   xT <- rbinom(1, n, p_treat)
5   # finish this: return the two-sided p-value comparing the two arms
6 }
7
8 # Run it 1000 times, count how many come back below 0.05, and plot the p-values.
```

# Two ways to turn 5% into much more

The 5% above assumed you ran **one** test, on **one** metric, and looked **once**.

Real teams do neither:

- They watch the dashboard daily and stop as soon as it goes green.
- They track conversion, revenue, retention, clicks, time-on-page, ... and report whichever moves.

Both feel harmless. Predict how far each one pushes the 5% before you measure it.



## Exercise 15 — Break it: peek, and test many metrics

Still a button that does **nothing**. Reuse `one_test()` from Exercise 14.

(a) **Peeking.** Instead of testing once at 2,000 visitors per arm, test after every 200 visitors and stop the moment any p-value dips below 0.05. Out of 1,000 such experiments, how often do you stop and declare a winner?

(b) **Many metrics.** Back to testing once, but on **10 independent metrics**, reporting a win if *any* of them is significant. How often does that happen? Compute the exact probability too.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 v one_test <- function(n = 2000, p_control = 0.06, p_treat = 0.06) {
2   xC <- rbinom(1, n, p_control)
3   xT <- rbinom(1, n, p_treat)
4   phat <- (xC + xT) / (2 * n)
5   z <- (xT/n - xC/n) / sqrt(phat * (1 - phat) * 2 / n)
6   2 * pnorm(-abs(z))
7 }
8
9 # (a) test after every 200 visitors, stop at the first p < 0.05
10
11 # (b) 10 metrics at once; also the exact probability
```

# How many visitors do you need?

Now flip it around. Suppose the new button genuinely lifts conversion from **6% to 7%** — a real effect, worth money.

Run the test on 2,000 visitors per arm and you might still miss it. The probability of *catching* a real effect is the test's **power**.

Guess the power at  $n = 2,000$  per arm. Then guess how large  $n$  has to be for a 90% chance of catching a 1-percentage-point lift.



## Exercise 16 — Sample size by simulation

Your guesses: power at  $n = 2,000$  is \_\_\_\_; the  $n$  needed for 90% power is \_\_\_\_.

Reuse `one_test()`, but now set the two arms to genuinely different rates.

1. For each  $n$  in 1,000 / 2,000 / 5,000 / 10,000 / 20,000 per arm, simulate 300 experiments at  $p_C = 0.06$  vs  $p_T = 0.07$  and record the share that reject at  $\alpha = 0.05$ .
2. Plot power against  $n$ .
3. Read off roughly what  $n$  buys you 90% power — and say what that means for how long the experiment must run if the site gets 5,000 visitors a day.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1_v one_test <- function(n, p_control, p_treat) {  
2   xC    <- rbinom(1, n, p_control)  
3   xT    <- rbinom(1, n, p_treat)  
4   phat  <- (xC + xT) / (2 * n)  
5   z     <- (xT/n - xC/n) / sqrt(phat * (1 - phat) * 2 / n)  
6   2 * pnorm(-abs(z))  
7 }  
8  
9 # Your code here
```

# Exam question — Fall 2025 Midterm 1

**Q16.** Design an **A/B test** to measure whether a change causes an increase in an outcome: define the treatment and control groups, the randomization, the outcome, and the two-sample test you would run.

We frame this in class using `Data/households_environment.rda` (object `exam_data`):

- **Randomize** units into treatment vs control → the groups are, on average, identical except for the treatment.
- Collect the **outcome**, then run a **two-sample difference-in-means test** ( $H_0 : \mu_T - \mu_C = 0$ ).

**Practice now:** the two-sample and A/B-design problems in the Week (Testing II) set.

→ [More exam practice at the end of this deck](#)

# Hypothesis Testing: Paired Data

# Paired data

- Sometimes we have only one set of individuals or objects, but two (or more) observations per each.
- **Example:** employee productivity — two observations per employee: productivity working from home and working from the office.
- **Example:** sales at Walmart and Soriana in each neighborhood — two observations per neighborhood.

# Paired data — setup

We have a set of independent pairs of data  $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ .

Observations within a pair are not necessarily independent!

- Some productive workers are productive both at home and in the office; some unproductive are unproductive in both.
- We observe the difference for each individual:  $X_i - Y_i = d_i$ .
- We look at the population value of that difference:  $\mu_X - \mu_Y = \Delta$ .
  - Check if the means are the same:  $\Delta = 0$  (no difference in productivity)
  - Or mean  $X$  is bigger than mean  $Y$ :  $\Delta > 0$

# Paired data — test statistic

Null hypothesis:  $H_0 : \Delta = \Delta_0$

Test Statistic is:

$$\text{test statistic} = \frac{\overbrace{\bar{X} - \bar{Y}}^{\bar{d}} - \Delta_0}{SE} = \frac{\bar{d} - \Delta_0}{\text{.....}SE\text{.....}}$$

- Note on the standard error — it's not the same formula as for two independent samples:

$$Var(X_i - Y_i) \neq Var(X_i) + Var(Y_i)$$

- We just calculate it from the differences:

- If we know the variance,  $SE = \frac{\sqrt{var(d_i)}}{\sqrt{n}} = \frac{\sigma_d}{\sqrt{n}}$

- If we don't:  $SE = \frac{s_d}{\sqrt{n}}$

# Paired data — rejection region

**Rejection Region:** for a test of significance level  $\alpha$ :

- If large sample or known variance with normal distribution:
- for  $H_A : \bar{d} \neq \Delta_0$ , reject if:

$$\frac{\bar{d} - \Delta_0}{SE} < z_{\frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{d} - \Delta_0}{SE} > z_{1-\frac{\alpha}{2}}$$

- for  $H_A : \bar{d} > \Delta_0$ , reject if:

$$\frac{\bar{d} - \Delta_0}{SE} > z_{1-\alpha}$$

- for  $H_A : \bar{d} < \Delta_0$ , reject if:

$$\frac{\bar{d} - \Delta_0}{SE} < z_{\alpha}$$

- If small sample from normal, replace critical values with the **t-distribution with n-1 degrees of freedom**.

# Paired or unpaired data?

1. We want to know if Intel's stock and Southwest Airlines' stock have similar rates of return. We take a random sample of 50 days and record both stocks on those same days.
  - Paired
2. We randomly sample 50 items at Target and note each price. Then we visit Walmart and collect the price for each of those same 50 items.
  - Paired
3. A school board wants to know if average SAT scores differ between two high schools. They take a simple random sample of 100 students from each school.
  - Not paired



## Exercise 17 — When does `paired = TRUE` apply?

Which of these is the condition that makes a t-test paired?

- the two samples have the same number of observations
- each unit contributes **one observation to each of the two measurements**
- the two samples come from the same population
- the outcome is continuous

**Each unit contributes one observation to each measurement** — the same employee, store or neighborhood measured twice. Then the two columns line up row by row, and you test the within-unit differences  $d_i$ .

Equal sample sizes are a *consequence* of pairing, not the cause: two independent samples can happen to have  $n = m$  and still be unpaired. If you can shuffle one column independently of the other without losing information, the data are not paired.

# Same 40 employees, measured twice

`productivity` holds 40 employees, each with an `office` score and a `home` score: the same person appears in both columns.

Some people are simply more productive than others — at home *and* at the office. That person-level variation is in both columns, and it is not what you are trying to measure.

The question is whether working from home changes output. Two ways to ask it:

- treat the 80 numbers as **two independent samples** of 40, or
- work with the 40 **differences** `home - office`.

Same 80 numbers. Should the two routes agree?



## Exercise 18 — Ignore the pairing, then use it

1. Look at the data: `head(productivity)`, and the correlation between the two columns.
2. Run the test **as if the two columns were independent samples**. Record the t statistic and the p-value.
3. Now build `d <- productivity$home - productivity$office` and run a **one-sample** test on `d` against 0. Record the same two numbers.
4. Confirm route 3 equals `t.test(..., paired = TRUE)`.
5. Compare `sd(productivity$home)`, `sd(productivity$office)` and `sd(d)`. Which one drives each standard error — and does that explain the gap between the two p-values?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

# Why pairing wins

- $Var(X_i - Y_i) = Var(X_i) + Var(Y_i) - 2 Cov(X_i, Y_i)$ .
- Positively correlated pairs  $\rightarrow$  that last term is **large and positive**  $\rightarrow$  the variance of the difference collapses.
- The independent-samples formula silently sets  $Cov = 0$  and inflates the SE.
- Cost: the paired test has  $n - 1$  df instead of roughly  $n + m - 2$ . Almost always a bargain.

# Exercise: Does IQ decrease with age?

A psychologist thinks age influences IQ. They take a random sample of 100 people aged 40. For each person we know their IQ at age 16 and now — two measurements on the same person.

On average, IQ at the young age was 8 points higher than at age 40, with a standard deviation of that difference of 7 points.

Using  $\alpha = 0.01$ , test the hypothesis that IQ decreases with age.



## Exercise 19 — The IQ test, from summaries

You are given only  $n$ ,  $\bar{d}$  and  $s_d$  — which is all a paired test needs.

1. Write  $H_0$  and  $H_A$  for “IQ decreases with age”, with  $d_i = \text{IQ}_{16} - \text{IQ}_{40}$ .
2. Compute the standard error and the test statistic.
3. Compute the p-value and the critical value at  $\alpha = 0.01$ .
4. State the decision, and say why this is a paired problem and not a two-sample one.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n    <- 100    # people
2 dbar <- 8      # mean of (IQ at 16 - IQ at 40)
3 s_d  <- 7      # sd of those differences
4
5 # your code here
```

# Hypothesis testing for correlation

**Example:** are cleanliness score (C) and price (P) correlated in the population?

Assume **normally distributed variables** in **independent pairs**  $\{(C_1, P_1), (C_2, P_2), \dots\}$ .

**Null hypothesis:**  $H_0 : \rho(C, P) = \rho_0$

**Test statistic** and its distribution under the null:

$$T_{test} = \frac{\hat{\rho}(C, P) - \rho_0}{\sqrt{\frac{1 - \hat{\rho}(C, P)^2}{n-2}}} \sim t_{n-2}$$

where  $\hat{\rho}(C, P)$  is the sample correlation coefficient.

**Alternatives and their rejection regions:**

- $H_A : \rho \neq \rho_0$  — reject if  $t_{test} > t_{1-\frac{\alpha}{2}, n-2}$  or  $t_{test} < t_{\frac{\alpha}{2}, n-2}$
- $H_A : \rho > \rho_0$  — reject if  $t_{test} > t_{1-\alpha, n-2}$
- $H_A : \rho < \rho_0$  — reject if  $t_{test} < t_{\alpha, n-2}$

# Are the assumptions plausible here?

The test above assumes: both variables **normally distributed**, pairs **independent**, and the relationship **linear**.

Those are assumptions about `review_scores_cleanliness` and `price` in our 200 listings — so look at them before trusting any p-value.

What would you have to draw to check each of the three?



# Exercise 20 — Check the assumptions

Using `Sample_list`:

1. Draw the distribution of `review_scores_cleanliness` and of `price`, side by side.
2. Draw the scatter plot of price against cleanliness score.
3. For each of the three assumptions, say whether the picture supports it — and if not, in which direction the violation runs.

Exercise

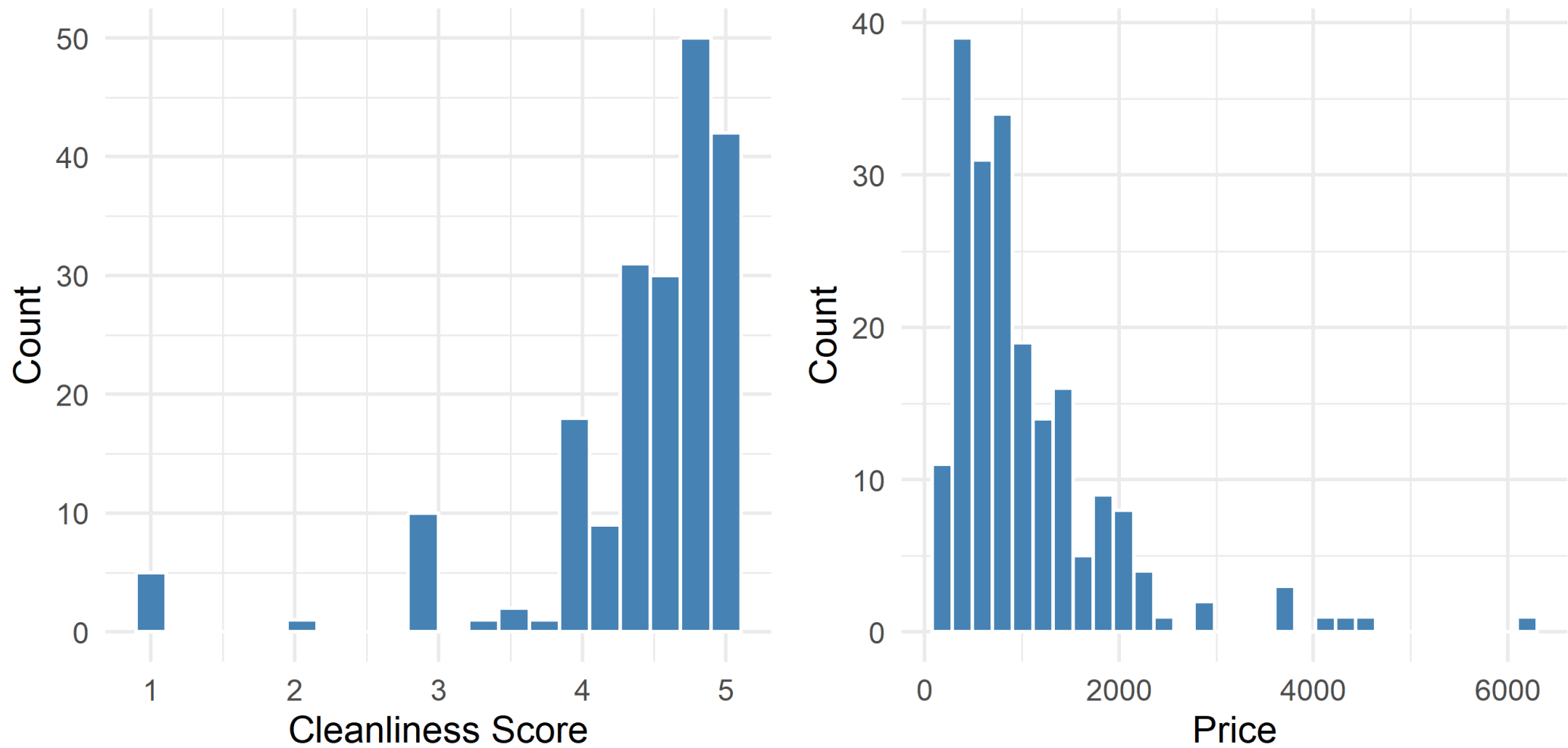
↺ Start Over

💡 Show Hint

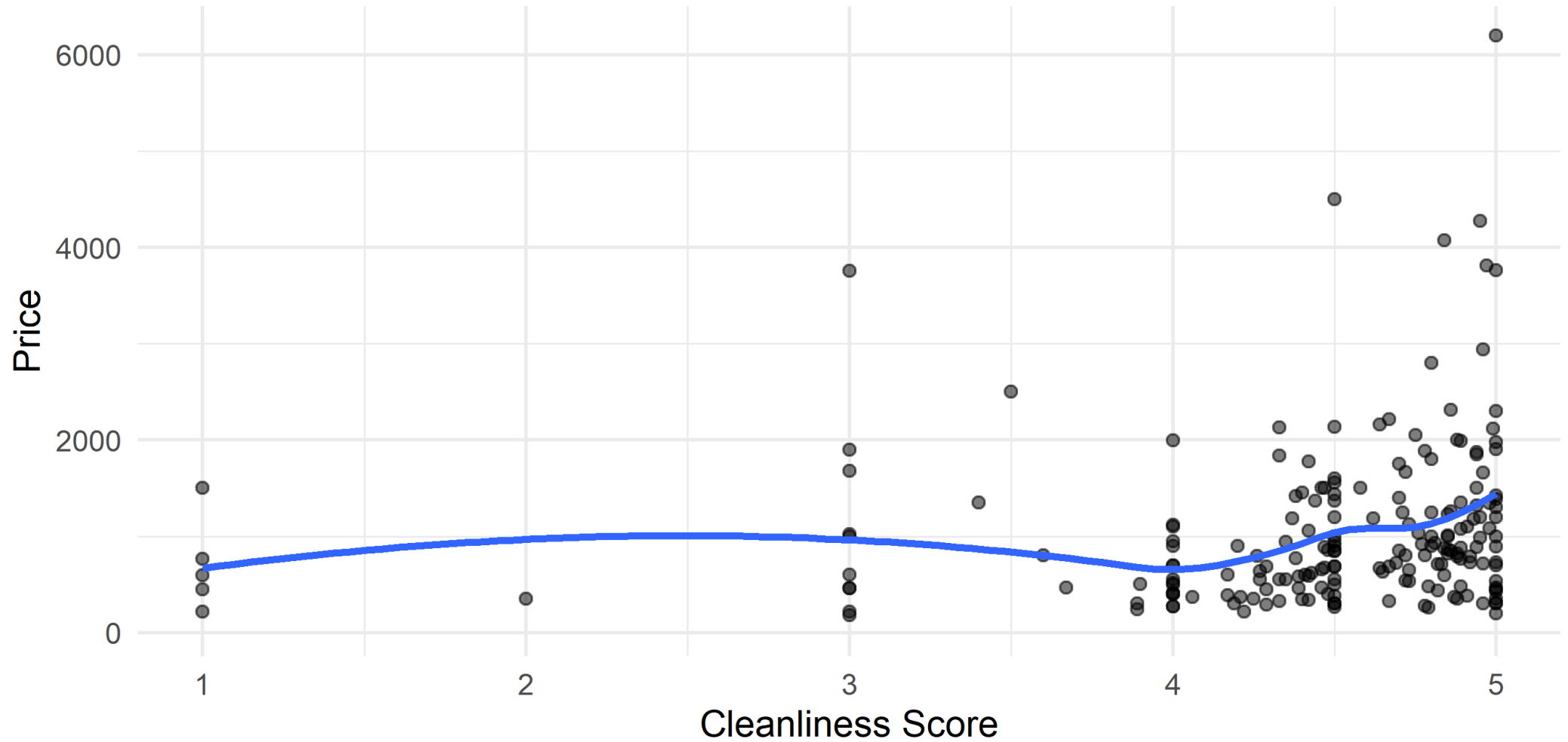
▶ Run Code

```
1 # Your code here
```

# Correlation test — distributions



# Correlation test — scatter plot



# Is the correlation different from zero?

We test  $H_0 : \rho(C, P) = 0$  against  $H_A : \rho(C, P) > 0$  — the one-sided version, because the business claim is that *cleaner* listings command *higher* prices.

$n = 200$ , so Student's t with 198 df is indistinguishable from the standard normal.

Compute  $\hat{\rho}$  and the statistic yourself before running `cor.test()`.

# Exercise 21 — Correlation test by hand, then by button

1. Compute the sample correlation between `review_scores_cleanliness` and `price`.
2. Build the test statistic from the formula  $t = \hat{\rho} \sqrt{\frac{n-2}{1-\hat{\rho}^2}}$ .
3. Compute the **one-sided** p-value and the critical value at  $\alpha = 0.05$ .
4. Check the whole thing with `cor.test()`.
5. The correlation is small. Is “statistically significant” the same as “big enough to act on”? Answer in one line.

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # Your code here
```

# Correlation test — the worked numbers

- We test  $H_0 : \rho(C, P) = 0$  vs  $\rho(C, P) > 0$  in our Airbnb data (clean score vs price).
- We **assume**: variables are normally distributed; pairs are independent; the relationship is linear.
- Sample correlation is 0.159, with  $n = 200$ .
- So  $t_{test} = \frac{\hat{\rho}\sqrt{(n-2)}}{\sqrt{1-\hat{\rho}^2}} = \frac{0.159\sqrt{198}}{\sqrt{1-0.025}} = 2.27$
- Since  $n = 200$ , Student's t is identical to the standard normal. Either compare to critical regions, or compute the p-value:
- $p\text{-value} = P(Z > t_{test}) = P(Z > 2.27) = 0.0126$
- A test at  $\alpha = 0.05$  would reject  $H_0$ .

# Exercise: Correlation test

Suppose you have a random sample of 27 units (from a bivariate normal).  $X$  measures a client's age and  $Y$  their spending. You calculated the correlation coefficient  $\hat{\rho} = -0.45$ .

Can you reject the null  $\rho = 0$  in favor of the alternative  $\rho \neq 0$  at the 5% significance level?



## Exercise 22 — Correlation with $n = 27$

Only two numbers are given, and that is enough.

1. Compute the test statistic and its degrees of freedom.
2. Compute the two-sided p-value and the two critical values at  $\alpha = 0.05$ .
3. State the decision.
4. Then ask a second question: with  $n = 27$ , how large would  $|\hat{\rho}|$  have to be to reach significance at all? Solve it by trying a grid of values.

Exercise ↺ Start Over 💡 Show Hint

▶ Run Code

```
1 rho <- -0.45
2 n   <- 27
3
4 # your code here
```

## Exercise 23 — The whole week, one question

You advise the Airbnb host portfolio. The owner asks:

*“We are considering paying for a deep-clean service on every listing. Is it worth it — and how sure are you?”*

Answer with a **number**, a **picture**, and **one sentence**. Then add the sentence the owner will not like.

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # 1. A number: how much more do clean listings earn, and how uncertain is that figure?
2
3 # 2. A picture: something the owner can read in three seconds.
4
5 # 3. One sentence for the owner.
6
7 # 4. One sentence on what this evidence CANNOT tell them.
```

# What you must be able to do this week

- **Two-sample difference in means:** build the test statistic, choose the SE and degrees of freedom, and reach a decision; construct a CI for the difference.
- **Paired data:** recognize when data are paired, reduce to differences  $d_i$ , and run the one-sample test on them — and know why that is more powerful.
- **Correlation test:** compute  $t_{test} = \hat{\rho} \sqrt{\frac{n-2}{1-\hat{\rho}^2}}$  and test  $H_0 : \rho = 0$ .
- **A/B testing:** design a randomized experiment, explain why randomization delivers **causal** inference, run the two-sample test on the outcome, and size the experiment before launching it.

**Course arc:** with testing done, the next weeks move from comparing groups to **modeling** relationships — regression.

# Exam practice

These problems mirror the Week 7 exercise set and the exam — work each one in the code cell, then press **Solution** to reveal the worked answer.

[← Back to the exam question slide](#)

# Exam practice 1 — Domestic vs international scores

Use the `metricsgrades` dataset (68 students in a graduate econometrics course). The key variable is `total`, a composite course score; the dataset also indicates whether each student is `domestic` or `international`. [I21]

Group summaries computed from the data:

| Group         | n  | Mean   | Variance |
|---------------|----|--------|----------|
| Domestic      | 32 | 77.719 | 50.002   |
| International | 36 | 76.236 | 67.332   |

(a) Provide an asymptotic **95% confidence interval** for the difference in mean total scores between domestic and international students. (b) Conduct a **z-test of equal means**. What is the p-value?

Exercise ↺ Start Over 💡 Show Hint ▶ Run Code

```
1 n_D <- 32; mean_D <- 77.719; var_D <- 50.002 # domestic
2 n_I <- 36; mean_I <- 76.236; var_I <- 67.332 # international
3
4 # (a) a 95% CI for the difference in mean total scores
5
6 # (b) the z-statistic and its two-sided p-value
```

Does the 95% CI contain 0 — and what do you conclude at  $\alpha = 0.05$ ?



# Exam practice 2 — Home Depot vs Lowe's returns

Use the [sp500](#) dataset, which includes monthly returns for several U.S. stocks and the overall market index. Key variables: [HD](#) (Home Depot returns), [LOW](#) (Lowe's returns), [IDX](#) (market index returns). [122]

Sample summaries ( $n = 364$  months): mean HD = 0.0167, var HD = 0.0054; mean LOW = 0.0203, var LOW = 0.0084.

(a) Compute the **z-statistic** and **p-value** for testing  $H_0 : \mu_{HD} = \mu_{LOW}$ , and state the decision at  $\alpha = 0.05$ .

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- 364
2 mean_HD <- 0.0167; var_HD <- 0.0054
3 mean_LOW <- 0.0203; var_LOW <- 0.0084
4
5 # the z-statistic and its two-sided p-value
```



## Exam practice 2 (cont.) — testing against a benchmark

(b) Suppose the market index mean return is 0.0078. Would it be appropriate to test  $H_0 : \mu_{HD} = 0.0078$ ?

First produce the number the question is really about: run the **one-sample** z-test of HD against that value, so you can see what it would claim.

Exercise ↺ Start Over 💡 Show Hint ▶ Run Code

```
1 n <- 364
2 mean_HD <- 0.0167; var_HD <- 0.0054
3 benchmark <- 0.0078
4
5 # the one-sample z-statistic and its two-sided p-value
```

Now the question itself: is that test appropriate?

**It depends on where 0.0078 comes from.** If it is an *external benchmark* (a historical average or theoretical prediction), treating it as a fixed constant in a one-sample test is fine.

But if 0.0078 is the *sample* mean of the index computed from the same data — and HD is a component of that index — then  $\bar{X}_{HD}$  and the benchmark are **dependent**, which violates the one-sample test's assumptions. The SE above is then wrong, and the p-value of 0.022 should not be trusted. A paired or regression-based approach would be more appropriate.



## Exam practice 3 — Wages and children: correlation test

Use the `cps` dataset (2809 employed individuals). Variables: `wagehr` (hourly wage in dollars), `ownchild` (number of children in the household). Test whether the **correlation** between wages and number of children is **zero**. [123]

State the null and alternative hypotheses.

$$H_0 : \rho = 0 \quad H_1 : \rho \neq 0$$

On the full data this is one call: `cor.test(cps$wagehr, cps$ownchild)`. It reports  $r = 0.0151$  over 1839 complete pairs. Rebuild the test statistic and p-value from those two numbers, then state the decision at  $\alpha = 0.05$ .

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 r <- 0.0151
2 n <- 1839      # complete pairs
3
4 # the test statistic, its degrees of freedom, and the two-sided p-value
```

At  $\alpha = 0.05$ , what do you conclude?

**Do NOT reject  $H_0$**  —  $r = 0.015$  is essentially zero and the p-value is 0.518. There is no statistically significant linear relationship between hourly wages and number of own children.